From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced)-2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
CRT–I
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**C**	**A, B, D**	**B**
2.	**B**	**A, D**	**A**
3.	**B**	**B, C, D**	**A**
4.	**B**	**A, D**	**D**
5.	**D**	**A, D**	**B, C, D**
6.	**C**	**B, C, D**	**A, B, C**
7.	**A**	**A, B**	**A, D**
8.	**C**	**B, C**	**A, B, C, D**
9.	**B.**	**D**	**C**
10.	**C**	**C**	**A**
11.	**B**	**A**	**C**
12.	**A**	**B**	**A**
13.	**C**	**B**	**A**
14.	**A**	**B**	**A**
15.	**A**	**D**	**A**
16.	**C**	**C**	**A**
17.	**B**	**A**	**C**
18.	**A**	**B**	**D**
19.	**C**	**D**	**B**
20.	**D**	**D**	**D**

Physics

PART – I

SECTION – A

2. $v_{max} = \omega \times mg/k$

$F_{av} = \dfrac{mv_{max}}{(T/4)} = \dfrac{2mg}{\pi}$

3. $W = \int_R^{2R}\left(p_0 + \dfrac{4S}{r}\right)4\pi r^2 dr = 4\pi p_0 \times \dfrac{7R^3}{3} + (4\pi)(4S) \times \dfrac{3R^2}{2}$

4. Work done by all the forces on the block equal to change in kinetic energy.

6. No effect of 'a' and 'g' on time period of spring pendulum.

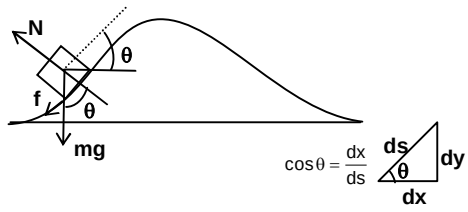
7. $\dfrac{1}{\lambda_{3-1}} = R\left(\dfrac{1}{1^2} - \dfrac{1}{3^2}\right) = \dfrac{8R}{9}$

$\dfrac{1}{\lambda_{2-1}} = R\left(\dfrac{1}{1^2} - \dfrac{1}{2^2}\right) = \dfrac{3R}{4}$

$\Rightarrow \dfrac{\lambda_{3-1}}{\lambda_{2-1}} = \dfrac{27}{32}$

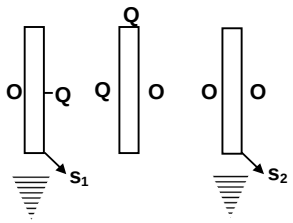
8. Work done by friction $= \int \vec{F}\cdot\vec{ds} = \int_0^x \mu mg\cos\theta \dfrac{dx}{\cos\theta}$

$= \mu mg\, x = 20$ J



9. Field inside the conductor is zero

10. Charge distribution is shown in figure.

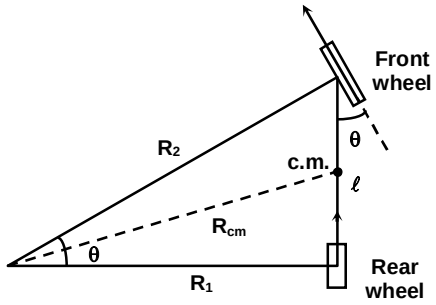


11. $\dfrac{\lambda}{(\sqrt{3mg})/(m/\ell)} = \dfrac{\lambda_0}{\sqrt{2mg/(m/\ell)}}$

$\Rightarrow \lambda = \sqrt{\dfrac{3}{2}}\lambda_0$

12. $v = \sqrt{\dfrac{\left(2m + \dfrac{m}{2}\right)g}{m/\ell}} = \sqrt{\dfrac{5}{2}g\ell}$

13-14. If the bike does not slip, path of both the wheel must be parallel with its direction as shown in the figure.



$R_1 = \ell/\tan\theta$

$R_{cm} = \sqrt{\left(\frac{\ell}{2}\right)^2 + R_1^2} = \frac{\ell}{2\tan\theta}\sqrt{4+\tan^2\theta}$

Net friction force $f = \frac{mv^2}{R_{cm}} = \frac{2mv^2\tan\theta}{\ell\sqrt{4+\tan^2\theta}}$

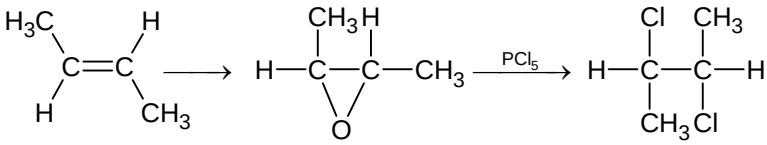
15. $p_0A = ma \Rightarrow a = p_0A/m$

16. $p_0A = m'a \Rightarrow m' = m/5$
$\therefore \quad m - m' = 4m/5$

Chemistry

PART – II

SECTION – A

2. 

(anti conformer)

3. Thermal stability of nitrates of alkali metals increases down the group. MgO is basic; SnO is amphoteric and B_2O_3 is acidic oxides.

4. Cell reaction is:
$Tl(s) + Cu^{2+}(aq) \rightleftharpoons Tl^{+}(aq) + Cu(s)$
Reaction quotient $Q = \frac{[Tl^+]}{[Cu^{2+}]}$ and Nernst equation is $E_{cell} = E^0_{cell} - \frac{0.0591}{n}\log Q$; at 298 K
To increase $E_{cell,}$ Q should be decreased. Which is decreased by decreasing $[Tl^+]$ and increasing $[Cu^{2+}]$.

5. Statement 'B' is true for weak electrolytes only and in case of 'C' osmosis never stops so dilution on solution side continues and no limiting value of equivalent conductance of solution is obtained.

6.

	Pb_3O_4 +	$4HNO_3 \longrightarrow$	$2Pb(NO_3)_2$ +	PbO_2	$+2H_2O$
Before reaction	0.05 mole	2 mole	–	–	
After reaction	–	1.8 mole	0.1 mole	0.05 mole residue	

$2Pb^{2+} + 8NaOH \longrightarrow 2Na_2[Pb(OH)_4]$
$HNO_3 + NaOH \longrightarrow NaNO_3 + H_2O$
Moles of NaOH used to neutralise excess of HNO_3 = 1.8 mole
Moles of NaOH used with $Pb(NO_3)_2$ = 0.4 mole
Total NaOH used = 2.2 mole.

7. Coagulation value $\propto \frac{1}{\text{Coagulating power}}$

Less is the gold number, better is the protection power.

8. Maximum capacity or volumes of balloon $= \frac{8}{7} \times 480 = 548.57$ ml

Also, $V_1 = 480$ ml, $T_1 = 278$ K, n = 1 mole

(A) The balloon will burst at a minimum temperature (T_2) when volume becomes 548.57 ml. Using Charle's law

$\therefore \frac{V_1}{T_1} = \frac{V_2}{T_2}$

$\frac{480}{278} = \frac{548.57}{T_2}$

$\therefore T_2 = 317.71$ K

or $T_2 = 44.71^oC$

Hence (A) is incorrect but (C) is correct.

(B) Pressure of the gas at 5^oC having 1 mole and V = 480 ml

PV = nRT

$\therefore P \times \frac{480}{1000} = 1 \times 0.0821 \times 278$

$\therefore$ P = 47.5 atm

Hence (B) is also correct.

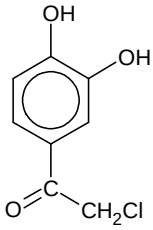
(D) Since V increase with increase in temperature from 5^oC to 44.71^oC, at which balloon bursts and therefore pressure remains constant. Thus pressure of gas inside the balloon when it bursts is 47.5 atm. Hence (D) is incorrect.

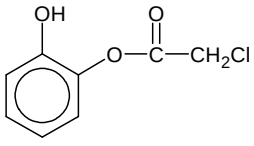
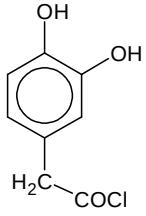
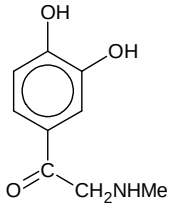
Hence (B, C) are correct.

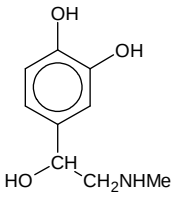
Solution for the Q. No. 9 & 10

B is $CO_2(g) \Rightarrow$ the compound A is a $ZnCO_3$ and C is oxide of Zn, as it is white when cold and turn yellow on heating. D is $CaCO_3$ which form $Ca(HCO_3)_2$ which exist in solution only.

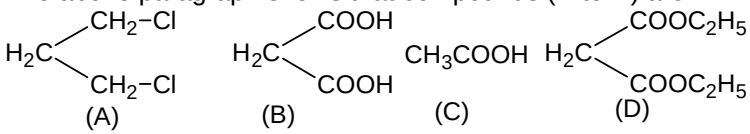
Solution for the Q. No. 11 & 12

Since 'A' react with NaOI, it must be  other side products are

 and less likely to be  ; B is  ;

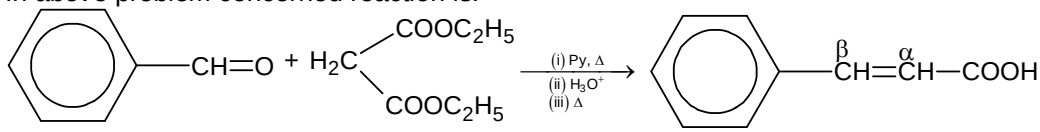
while 'C' is 

13. The above paragraph shows that compounds (A to D) are



(A) (B) (C) (D)

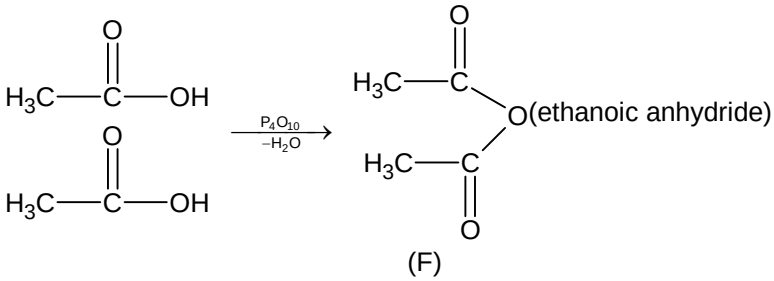
In above problem concerned reaction is:



Benzaldehyde (D) Cinnamic acid (E)

Hence (B) is correct.

14. Since compound (C) is CH_3COOH. So,



(F)

Hence (B) is correct.

15. (A) That is path independent e.g. $\Delta U = q + w$; U is state function, while q and w are path functions.
(B) Work is a type of energy in transit, i.e. which appears only on surface/boundaries.
(C) w & q are not completely interconvertible.

16. (A) Work is path function.
(B) Initially there is no opposing force but opposing force increases gradually due to diffused gas.
(C) $W = \pi ab$; (area inside the curve)
a = 3 litre, b = 1 atm.

17. Le-chateliers principle is applicable and for (R) $P_{total} = P_{CO_2}$, which is equal to K_p for the given reaction.

20. Isodiaphers are those atoms for which (n-p) is equal. Isoesters are those species which contains equal number of atoms as well as equal number of electrons.

Mathematics

PART – III

SECTION – A

1. Normal vector to plane P_1 $3\hat{k}$

Normal vector to plane P_2 $2\hat{i} - 4\hat{j} - 3\hat{k}$

$\vec{a} = \lambda\left(2\hat{i} + \hat{j}\right)$

angle between $\vec{a}$ and vector $\hat{i} - 2\hat{j} + 2\hat{k}$ is given by

$$\cos\theta = \frac{\lambda\left(2\hat{i} + \hat{j}\right)\cdot\left(\hat{i} - 2\hat{j} + 2\hat{k}\right)}{\lambda\sqrt{5}\sqrt{9}} = 0$$

$$\Rightarrow \theta = \frac{\pi}{2}.$$

2. $f'(x) = f(x)$

Integrating $\log f(x) = x + k$

$f(x) = e^{x+k}$

$f(0) = 1$

$1 = e^0.e^k$

$k = 0$

$\therefore f(x) = e^x$

$g(x) = x - f(x) = x - e^x$

$$I = \int_0^1 e^x\left(x - e^x\right)dx = \int_0^1 e^x x\,dx - \int_0^1 e^{2x}dx$$

$$= e - (e-1) - \frac{e^2}{2} + \frac{1}{2} = \frac{3}{2} - \frac{e^2}{2} = \frac{3 - e^2}{2}.$$

3. $n(S) = {}^{64}C_3$

Let 'E' be the even of selecting 3 squares which from the letter 'L'

No ways of selecting squares consisting of 4 unit squares = ${}^7C_1 \times {}^7C_1 = 49$

Each square with 4 unit squares forms 4L – shapes consisting of 3 unit squares

$\therefore n(E) = 7 \times 7 \times 4 = 196$

$$\therefore p(E) = \frac{196}{{}^{64}C_3}$$

4. $x^2 + 9y^2 + 25z^2 = 15yz + 5zx + 3xy$

$(x)^2 + (3y)^2 + (5z)^2 - (x)(3y) - (3y)(5z) - (x)(5z) = 0$

$$\frac{1}{2}\left[2(x)^2 + 2(3y)^2 + 2(5z)^2 - 2(x)(3y) - 2(3y)(5z) - 2(x)(5z)\right]$$

$$= \frac{1}{2}\left[(x-3y)^2 + (3y-5z)^2 + (x-5z)^2\right] = 0$$

$x - 3y = 0,\ 3y - 5z = 0,\ x - 5z = 0$

$$\frac{1}{x} = \frac{1}{3y} \text{ and } \frac{5}{3y} = \frac{1}{z} \text{ and } \frac{1}{5z} = \frac{1}{x}$$

$$\frac{1}{x} + \frac{1}{z} = \frac{1}{3y} + \frac{5}{3y} = \frac{2}{y}$$

$\therefore \frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P. and x, y, z are in H.P.

5. When z = n + 1 we can choose x, y from {1, 2, …, n}
$\therefore$ when z = n + 1, x, y can be chosen in n^2 ways and z = n, x, y can be chosen in $(n-1)^2$ ways and so on

$\therefore n^2 + (n-1)^2 + \ldots + 1^2 = \frac{1}{6}n(n+1)(2n+1)$

ways of choosing triplets
Alternatively triplets with x = y < z, x < y < z, y < x < z can be chosen in
$^{n+1}C_2$, $^{n+1}C_3$, $^{n+1}C_3$ ways.
There are
$^{n+1}C_2 + 2(^{n+1}C_3) = {}^{n+2}C_2 + {}^{n+1}C_3 = 2(^{n+2}C_3) - {}^{n+1}C_2$.

6. $z' = ze^{i\alpha}$ … (1)
$z'' = ze^{-i\alpha}$ … (2)
$\therefore z'z'' = z^2 \Rightarrow z', z, z''$ are in G.P.

$$\left(\frac{z'}{z}\right)^2 + \left(\frac{z''}{z}\right)^2 = 2\cos 2\alpha$$

$\Rightarrow z'^2 + z''^2 = 2z^2 \cos 2\alpha$
$z' + z'' = 2z\cos\alpha$.

7. $$\left|\frac{\frac{c}{3}}{\sqrt{\frac{1}{3^2}+\frac{1}{4^2}}}\right| = c \Rightarrow c = 6, 1.$$

8. $\cos\alpha = \frac{1}{2}\left(x + \frac{1}{x}\right)$ and $\cos\beta = \frac{1}{2}\left(y + \frac{1}{y}\right)$

since $xy > 0 \Rightarrow x + \frac{1}{x} \geq 2$ or ≤ -2

$y + \frac{1}{y} \geq 2$ or ≤ -2

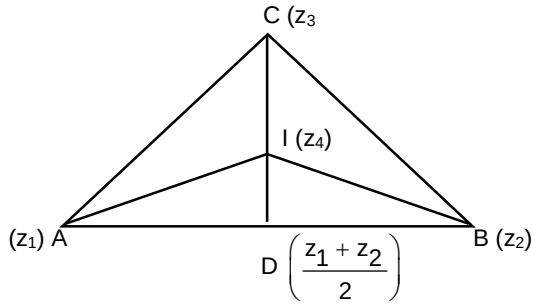
$\Rightarrow \cos\alpha = 1, \cos\beta = 1$ …(1)
or
$\cos\alpha = -1, \cos\beta = -1$ …(2)
$\therefore \cos\alpha\cos\beta = 1$
$(\cos\alpha + \cos\beta)^2 = 4$
$\Rightarrow \alpha + \beta$ is an even multiple of π.
$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin(2n\pi + \gamma) = \sin\gamma$
Also, $\sin\alpha = \sin\beta = 0$

9. $\angle IAB = \frac{\theta}{2}$, $\angle IAC = \frac{\theta}{2}$

$$\frac{z_2 - z_1}{|z_2 - z_1|} = \frac{z_4 - z_1}{|z_4 - z_1|}e^{-\frac{i\theta}{2}}$$

$$\frac{z_3 - z_1}{|z_3 - z_1|} = \frac{z_4 - z_1}{|z_2 - z_1|}e^{-\frac{i\theta}{2}}$$

$$\frac{(z_2 - z_1)(z_3 - z_1)}{|z_2 - z_1||z_3 - z_1|} = \frac{(z_4 - z_1)^2}{|z_4 - z_1|^2}e^{\theta}$$



$\therefore \dfrac{(z_2 - z_1)(z_3 - z_1)}{(z_4 - z_1)^2} = \dfrac{AB \cdot AC}{(IA)^2} = \left(\dfrac{AB}{IA}\right)^2 \left(\dfrac{AC}{AB}\right)$

10. $\dfrac{(z_2 - z_1)(z_3 - z_1)}{(z_4 - z_1)^2} = 2\left(\dfrac{AD}{IA}\right)^2 \left(\dfrac{AC}{AD}\right)$ (Since AB = 2AD)

$(z_4 - z_1)^2 (1 + \cos\theta) \sec\theta = (z_2 - z_1)(z_3 - z_1)$

11. Equation of the tangent is

$\dfrac{x}{a}\sec\theta - \dfrac{y}{b}\tan\theta = 1$

Normal is $ax \cos\theta + by \cot\theta = a^2 + b^2$

The normal at P meets the co-ordinate axes at $G\left(\dfrac{a^2+b^2}{a}\sec\theta,\ 0\right)$ and $g\left(0,\ \dfrac{a^2+b^2}{a}\tan\theta\right)$

$\therefore PG^2 = \left(\dfrac{a^2+b^2}{a}\sec\theta - a\sec\theta\right)^2 + (b\tan\theta - 0)^2$

$PG^2 = \dfrac{b^2}{a^2}(b^2\sec^2\theta + a^2\tan^2\theta)$

12. When $\tan\theta = 0$

$PG = \dfrac{b^2}{a}$

13. When 1 all rounder and 10 players from bowlers and batsman play number of ways
$= {}^4C_1 \cdot {}^{14}C_{10}$
when 1 wicket keeper and 10 players from bowlers and batsman play number of ways
$= {}^2C_1 \cdot {}^{14}C_{10}$
when 1 all rounder 1 wicket keeper and 9 from batsmen and bowlers play number of ways
$= {}^4C_1 \cdot {}^2C_1 \cdot {}^{14}C_9$
when all eleven players play from bowlers and batsmen then the number of ways
$= {}^{14}C_{11}$
$\therefore$ Total number of selections $= {}^4C_1 \cdot {}^{14}C_{10} + {}^2C_1 \cdot {}^{14}C_{10} + {}^4C_1 \cdot {}^2C_1 \cdot {}^{14}C_9 + {}^{14}C_{11}$.

14. If 2 batsmen don't want to play then the rest of 10 players can be selected from 17 other players number of selection $= {}^{17}C_{10}$
If the particular bowler doesn't play then number of selection $= {}^{19}C_{11}$
$\therefore$ Total number of selection $= {}^{17}C_{10} + {}^{19}C_{11}$

15. Slope of tangent at any point P(x, y) is $\dfrac{dy}{dx} \propto y$

$\Rightarrow \dfrac{dy}{dx} = ky \Rightarrow y = e^{k(x-1)}$

Since $a(y - 1) + (x - 1) = 0$ is normal at $(1, 1) \Rightarrow k = 1$
Hence $f(x) = y = e^{a(x-1)}$

16. Area bounded $= \displaystyle\int_0^1 \left(\frac{1}{a} - \frac{x}{a} + 1 - e^{a(x-1)}\right)dx = \frac{1}{a}\left(a - \frac{1}{2} + e^{-a}\right)$ sq. units.

17. (P) Limit is easily reducible to $\operatorname{cosec}^{-1}\left(\frac{k^2}{2}\right)$ by L'Hospital rule which exist if $\frac{k^2}{2} \geq 1$.

(Q) $kx^2 + (3 - 2k)x - 6 = (kx + 3)(kx - 2)$

$4 \leq -\frac{3}{k} \leq 5 \Rightarrow -\frac{3}{4} \leq k \leq -\frac{3}{5}$.

(R) $(2k + 1, k - 1)$ is an interior point

$(2k + 1)^2 + (k - 1)^2 - 2(2k + 1) - 4(k - 1) - 4 < 0$

$0 < k < \frac{6}{5}$ ..... (1)

Centre (1, 2) and point $(2k + 1, k - 1)$ must lie on opposite side of chord $x + y - z = 0$

$k < \frac{2}{3}$ ..... (2)

$0 < k < \frac{2}{3}$

(T) $x > -\frac{5}{2}, -4 < x < 4 \Rightarrow x \in \left(-\frac{5}{2}, 4\right)$

$\log_5\left(\frac{16 - x^2}{2x + 5}\right) \leq 1$

$\Rightarrow \frac{16 - x^2}{2x + 5} \leq 5^1$

$\Rightarrow x \in (-\infty, -9) \cup [-1, \infty)$

$\therefore x \in [-1, 4]$

18. $I = \int_0^\infty e^{-tx^2} dt$

$x\sqrt{t} = u$, then $dx = \frac{du}{\sqrt{t}}$

$I = \int_0^\infty e^{-u^2} \frac{du}{\sqrt{t}} = \frac{1}{\sqrt{t}} \int_0^\infty e^{-u^2} du = \frac{\sqrt{\pi}}{2\sqrt{t}}$

$\int_{-\infty}^{\infty} e^{-x^2} dx = 2\int_0^\infty e^{-x^2} dx = \sqrt{\pi}$.

19. $r_1r_2 + r_2r_3 + r_3r_1 = \frac{r_1r_2r_3}{r} = s^2 = 144$

$4(r_2 + r_3) + r_2r_3 = 2r_2r_3 = 144$

$\Rightarrow r_2r_3 = 72, r_2 + r_3 = 18 \Rightarrow r_2 = 6, r_3 = 12$

$\frac{r}{r_1} = \frac{s - a}{s} = \frac{1}{2} \Rightarrow a = 6$

Similarly $b = 8, c = 10$

$\Rightarrow$ ABC is right angle triangle.

Smallest angle is $\sin^{-1}\frac{3}{4}$

$\Delta = \frac{1}{2} \times 6 \times 8 = 24$ sq. units

$R = 5$

$|r_2 - r_3| = 6$

20. (P) Equation of normal is $y = -tx + 2at + at^3$ at P(t)
It intersect the curve again at point $Q(t_1)$ on the parabola such that

$$t_2 = -t - \frac{2}{t}$$

Again slope of OP is $\frac{2}{t} = M_{OP}$

Also, slope of OQ is $\frac{2}{t_1} = M_{OQ}$

Since $M_{OP} \cdot M_{OQ} = -1 = \frac{4}{tt_1}$

$\Rightarrow tt_1 = -4$

$$t\left(-t - \frac{2}{t}\right) = -4$$

$\Rightarrow t^2 = 2$

(Q) P(1, 2), Q(4, 4), R(16, 8)
Now, ar(ΔPQR) = 6 sq. units

(R) Equation of normal from any point $P(am^2, -2m)$ is
$y = mx - 2am - am^3$
It passes through $\left(\frac{11}{4}, \frac{1}{4}\right)$
$\Rightarrow 4m^3 + 8m - 11m + 1 = 0$
$\Rightarrow 4m^3 - 3m + 1 = 0$
Now, $f(m) = 4m^3 - 3m$
$\Rightarrow f'(m) = 12m^2 - 3 = 0$

$$\Rightarrow m \pm \frac{1}{2}$$

Since $f\left(\frac{1}{2}\right) f\left(-\frac{1}{2}\right) < 0$ has 3 normals are possible.

(T) Since, normal at $P(t_1)$ if meets the curve again at (t_2), then

$$t_2 = -t_1 - \frac{2}{t_1}$$

Such that here normal at P(1) meets the curve again at Q(t)

$$\Rightarrow t = -1 - \frac{1}{2} = -3$$